

PID	Search Anchoring	Difficulties	Data Quality Issues	Is Determinative	Tools Used	Information Access Methods	Purchasing Decision	Time	Open code	Quote (optional)	Notes (optional)
P28								10:37:53	try to visualize a subset and the entire dataset to choose the most interesting ones	Because how long of a time period you wanna include the data for is always a question visualizations given a certain range	
P32								0:06:29	poster that visualize how the human force is involved into AI training data set		
P32								0:07:55	asked 3 chatbots to get the dataset		
P32	D, P							0:08:09	got gave a seemingly real dataset at the first trial, but the next trial with the same prompt gave fake data (with descriptions suggest they did not have access to real data)		
P32								0:09:46	usually search through official organizations		
P32						P		0:16:33	profiling connected with users		
P32						D		0:16:48	reputable source, documentation on how the data has been collected, enough metadata		
P32								0:17:14	don't care about usage metrics		
P32								0:17:21	data labels' consistency, otherwise it requires more attention	I find a valuable when data and terms and labels are written correctly across all the maybe it's a database or whatever	
P32		Y				V		0:17:59	accompanying visualizations help get a quick glimpse of the data	I think about the times when I go on eurolast which is in largest statistical database for Europe. Like having a short table with some ready to made graphs. Really help me to get a glimpse of what's inside.	
P32						F		0:18:23	also check when the data was published or updated		
P32								0:18:33	usually don't check ratings: probably due to not participating in any competition or hackathon		
P32							Y	0:18:58	access restriction: if I need to buy the data, I need to know what's inside		
P32						D		0:19:50	metadata includes the definitions of the columns and cell values, data collection time	Taking Eurolast as an example, they always have a secondary table with all the definitions of the string values or items, etc. And they usually writes how these have been captured. What's the last time that these have been collected	
P32						D		0:20:23	such details help data scientists, designers, journalists get extra details, but	So these kind of small details help the designer or the data scientist to really be extra sure about the details. To be honest, I have never worked on a project in which such details was required.	
P32								0:21:08	consider metadata as a part of documentation	I assume that under documentation I will find metadata	metadata is included in documentation
P32		Y				T		0:22:02	visual hierarchy is important: use life and description first		
P32								0:22:21	sample data: need a preview of the data		
P32								0:22:21	get a sense of how many columns to display		
P32						V, P		0:22:40	want charts that visualize the combinations of some variables		
P32		Y			Y	P		0:22:52	otherwise, would download the data and figure out themselves with other tools		
P32								0:23:06	if the data is important, then look into the documentation		
P32								0:23:16	never need to contact the data provider, because mostly work on personal projects, or datasets approved by manager, or datasets brought by clients		not taking much responsibility in terms of choosing the right dataset
P32						F		0:23:44	usually provide data on when the data's been collected, so need such information		
P32		Y				S		0:25:32	if dataset is too large, then download sample data		
P32								0:26:24	part is about creating visualization to see what the how much this data set is valuable		
P32								0:26:34	as a designer, always think visually		
P32						V		0:27:17	but those visualizations are always while exploring the dataset	I guess this is because I'm a designer. And I really think visually, so I don't think I'm skilled enough to just by looking at the data to get an idea if this is the right one or not. But usually I download multiple data sets, and then I test visually which of these is more adapt to my case, or which one is has more information, and which one is richer or n	
P32								0:28:00	suggested tools used: row graphs, flourish that are wrapper, map box, and the tableau		
P32		Y		Y				0:28:08	suggested tableau is powerful but hard to customize	I know tableau is the most powerful. It's not super super easy to customize, and I mean technical background matters	
P32								0:29:05	may use multiple tools, also use: chart master inside the illustrator and also some plug-in a bit of a mix. It depends on how difficult is the chart, how big is the data set.		
P32								0:30:12	sometimes even google sheet or excel, but those are just intermediate products, never the final results		
P32								0:30:26	use these two to sketch but options are usually too limited	I sketch a lot about the visualization, I usually think about the visualization types and which one can fit. The type of data that I have, and most of the times the option is not available on the Google spreadsheet or excel	
P32								0:30:54	unless it's a report need simple graphs		
P32		Y	Y	Y				0:31:40	asked what is the definition of data quality – more towards this definition: how the data has been collect. If there is metadata, and if everything is downloadable by the user.		
P32			Y					0:32:01	on the definition could be formatted in the right way, ready to use	maybe they just show you what happened, and I don't know it one month or the summary of the year. So you don't really get stuck at bar chart or bar chart divided by quarters and so on.	
P32			Y	Y		D, F		0:32:37	data quality relates to documentation, metadata, etc. and the year of publishing or update		
P32			Y					0:32:50	often use data in pdf, which is usually not clean but still valuable	I often use data that it's like, you know, formatted as a Pdf, or maybe it's not clean, but nonetheless it's still valuable. So, of course, takes a little bit to manually clean it or through other tools. But it's not that I value it as less just because it's not perfect.	
P32			Y	Y				0:33:05	wouldn't value a dataset that requires cleaning to be less valuable		
P32								0:34:02	data quality has to do with whether is actual data or fake data		
P32			Y	Y		F		0:34:50	data quality issue: cannot be combined with other data due to the time it been collected: need 2020 but got 2016		
P32			Y	Y				0:35:53	how things were in 2016 and 2020 might be different, so not relevant to logic		
P32							Y	0:36:34	never purchased		
P32							Y	0:36:59	data for political census or opinions or cultural mindset: the kind of data that marketing really wants should be expensive because the organize who wants it can make money out of it and can afford it		
P32							Y	0:37:28	data about things that citizens would benefit from getting it should be affordable to most people		
P32							Y	0:41:39	I really need it and within budget, otherwise go for alternatives		
P32						V		0:43:19	bar charts, pie charts and line graphs	I always see bar charts, pie charts and line graphs.	
P32		Y		Y		V		0:43:33	hard to compare	Pine charts has more than 4 or 5 values displayed, so it becomes very difficult (to compare)	
P32								0:43:28	selected items are not useful, because the automatically generated ones include m	Maybe the bar charts. They show just a fraction of the data set.	
P32						V		0:43:41	sometimes doesn't give summarized information	maybe they just show you what happened, and I don't know it one month or the summary of the year. So you don't really get stuck at bar chart or bar chart divided by quarters and so on.	
P32		Y		Y		V		0:43:56	the selected trend is not an important information but grabs attention, which leads to and the line graphs. They are very common, but they're usually used very randomly. So when you check the line graph. And if you're not used to working with data, you might really think that. Maybe this data set is not very useful, but it's only because they picked one specific trend		
P32								0:44:19	the combination doesn't make sense	Or maybe it's a combination of data that doesn't make sense when represented with the line graphs.	
P32						V		0:44:19	difficult to provide visualizations of all possible combinations	So obviously, it's difficult to making available every type of charts to in any case.	
P32						V		0:44:49	provide warning message about the limitation of the provided visualization may be h	Maybe it would be helpful to make better that the chart is only representing a fraction of the data, so make it very clear in the description or in the title.	
P32						V		0:46:22	always try to show relationship among data		
P32						V		0:46:29	quantity corresponds to bar charts		
P32						V		0:46:36	mentioned pie charts		
P32						V		0:46:46	also try tree map (hierarchical chart)		
P32						V		0:47:09	always use visualization to show relationships among columns	it's not really like, show the bigger picture of the data set through visualizations, but trying to make it clear that these data sets are not on their own, it/they're always related to certain parameters depending by the story you display	
P32						V		0:47:17	use a lot of diagrams like bees worm plot, bubble charts to show dimensions		
P32						V		0:47:46	always a combined effort among 2 or more data charts	the provided visualizations can also be a combined effort	
P32						V		0:49:26	group things first by time or item then play around with visualizations		
P32						V		0:52:13	combine effort: use one chart to show the region and one chart to show the value		
P32						V, P		0:53:45	sometimes the relationship between variables is clear, sometimes is not, try around		
P38						P		7:45:05	post-graduate project: choose certain topic, use visualization or other data exploration to see if it's valid, then do some analysis		
P38	P							7:46:14	first go to the authoritative open source/data platform		
P38								7:46:30	the data they look for was some specific: geospatial temporal data		
P38	P							7:46:37	goes to several well-known big platforms		
P38		Y						7:47:01	found some basic information but not the specific required dataset		
P38	D							7:47:43	searched works related to the chosen topic on google scholar and found the datasets these works referred to		
P38	D, P							7:48:54	learned more platform and resources where I could get data for free		
P38								7:49:11	then explored to see if these datasets works	if I could deliver more information based on the messad up the, Jonn one, but it will be very time consuming.	
P38						R		7:49:28	as those datasets are referred by previous works, they are more likely to be reliable		
P38								7:50:31	for working scenarios, have internal data platform holding many datasets created by different departments, archive frequently used datasets		
P38		Y						7:51:23	have access to many data, but unsure about the data collection conditions, usage sc	I could get access to lots of lots of different kinds of dataset, but I always have the question of: How do they get those data? And what's the upgrade frequency? And are those data cover all different scenarios? what's the trigger conditions?	
P38						D, C		7:51:51	I know details about datasets directly relevant to my work scope, but less familiar w	I focus on my work scope. I will know lots of details of the data says I'm relevant to. But for other data, I really need to ask the data set owner, or to maybe set up meetings to discuss more about those. I would say the data sets in my in my company is not that developed.	
P38			Y			D, C		7:52:17	datasets is not developed though have lots of data, the maintenance is bad. Need k	I would say the data sets in my in my company is not that developed. Even we have the profound / cross use case	
P38								7:53:12	'bigger scenario is when to record specific events in the video stream: such as the clicks		
P38								7:56:23	Structure of the internal platform: use case/function as a "database"-many datasets-several tables		
P38	D							7:58:12	for search, still do keywords search: can match both use cases and datasets that contain the keywords		
P38	D							7:59:23	can use dataset ID / knowing it		
P38		Y						7:58:41	need improvements: the keyword search results sometimes are not satisfying: name Even I use the correct keywords. The results might not be that satisfying, or that relevant to my requirements. Cause as I've mentioned before, the maintenance of those assets. The names are very out of because sometimes needed on the data set for themselves for their department to use. So there's no need to do that much maintenance th		
P38						S	Y	8:03:09	I doing academic research, I want to export the data for exploration and see the cost		
P38						V		8:03:47	for academic research specifically, providing visualization functions or having visualization inside the website adding its persuasiveness	compared with academic research: how information being presented is important	
P38						D, A, C		8:04:34	but in workplace, first row, especially documentation is more important		
P38						D, F, T, U		8:04:52	documentation contains the origin and all the details and many icons on our list: like the update frequency, attributes, possibilities in the future, and sample data		
P38		Y						8:05:57	if I need to use the data to fulfill my work, I need to avoid the risk		
P38		Y						8:06:34	ratings and reviews might be more suitable for the academic scenario: if I see people's comments on its robustness or see it being used in academic articles, I'm more likely to use the dataset		
P38						R		8:07:09	ratings and reviews in workplace are more subjective and case-specific	expectations	
P38						V, D		8:07:46	for academic research, presenting method is important, but in the workplace, documentations is more important due to efficiency		
P38						U		8:08:44	future possibilities are tasks that may rely on this dataset for follow up analysis		
P38								8:09:54	internal platform doesn't provide visualizations		
P38						V		8:10:12	the internal platform currently pays more attention to data than the presentation, but So actually, you give me a hint. If in the future, we want to improve that data internal platform, we may make use of a realization tool.		
P38		Y						8:12:09	I given two datasets: one has more clear attributes, and the other contains more useful information	an interesting example	
P38								8:14:40	need to think about what's the purpose of using that dataset	if I could deliver more information based on the messad up the, Jonn one, but it will be very time consuming.	
P38								8:15:30	in working scenario, using the first one might increase efficiency	that's something from working scenario to increase the efficiency and to make everyone pleasant. I will just use the 1st data sets.	
P38		Y						8:17:51	don't go for the documentation when I made the decisions. I think when I go to document contradicts themselves		
P38						S		8:18:14	for research, would implement both on model to check performance or see which one is easier to interpret		
P38		Y						8:18:50	in workplace, would ask colleagues and follow their trace, especially if need to collaborate in the future		extra constraints/advice when collaborate
P38								8:19:23	for research, wouldn't think much about others' preferences		
P38						T, S		8:20:05	if new to data, will search for sample data first to get a glimpse		
P38						F, X		8:20:13	check data size and update time		
P38						T, D		8:20:26	some attributes might be helpful, but some might not have a straightforward interpretations, then go to documentation		
P38								8:21:44	in workplace, the robustness of the data is usually good, captures enough data points		
P38		Y	Y	Y				8:22:09	for academic research where searched data online, two datasets with different v	I search the data through websites or some platforms, sometimes I compare different v	you may find there will be deviation in different data sets, and you don't know which one you should rely on.

PID	Search	Anchor	Difficulties	Data Quality Issues	Is Determinative	Tools Used	Information Access Methods	Purchasing Decision	Time	Open code	Quote (optional)	Notes (optional)
P38		Y	Y	Y						8:23:27 different attributes associated to the same measurement (timestamp in two time zones), confused about which one to use		
P38		Y	Y	Y						8:23:52 also confused about the data collection process: e.g. if it's in another country, then does the measurement use Chinese time?		
P38		Y	Y	Y			D			8:24:22 need check documentation for clarification: sometimes even missing		
P38		Y	Y	Y						8:24:53 different datasets use different collecting frequency		
P38		Y	Y	Y			F			8:27:00 sufficient: minimal missing data, need data to be new and complete		
P38		Y	Y	Y						8:27:22 datasets across platform should be the same volume		
P38		Y	Y	Y						8:27:49 enough data if the data is the same, across different platforms		
P38		Y	Y	Y						8:28:23 attribute is easy to interpret		
P38		Y	Y	Y						8:28:42 for workplace only: been separated clearly (cleaning)		
P38		Y	Y	Y						8:29:03 because don't want to spend time doing verification		
P38		Y	Y	Y			S, T			8:29:53 check sample data to see if it makes sense		
P38		Y	Y	Y						8:30:19 contact data owner to fix major issues		
P38								Y		8:31:16 old purchase		
P38								Y		8:32:14 no alternative, all such data are provided by the same provider; reasonable price		
P38							V, P			8:35:56 visualize data volum, distribution		
P38							V, U			8:36:13 for data sharing platform, maybe use visualization to show insights/basic possibilities that can be drawn from the dataset		
P38							V			8:37:28 map can be useful		
P38							V			8:37:36 visually pleasing visualization can attract more users; they may feel they can generalize Charming Colorful way to present their data using visualization methods. It's might be more attractive to users. Will help them think that they may generate the same thing by using their data.		
P38							V, P			8:39:41 visualization for insights: relationship between two variables, 3 vars have high correlation, high significance with the results		
P38										15:39:42 fraud detection for credit card		
P38										15:40:54 company's internal platform		
P38										15:42:11 build ML model using features like transaction details		
P38							C			15:42:53 selected based on peer's experience or just throw them in		
P38										15:45:43 work on a stable model, doesn't have to change many features		
P38										15:46:08 add new features according to performance		
P38										15:49:10 internal platform hold many separate tables with data dictionary		
P38		Y		Y			U			15:49:21 need to know the meaning of each feature, otherwise wouldn't use	you definitely need to have a description like what's the meaning of these features. That's really important for us because if you don't have this description, people will not know, like what's the meaning of the feature and many people just leave it there.	
P38							U, R			15:51:57 use case and user reviews are helpful		
P38										15:52:01 get a sense of how people used the data and a quick, general picture of the dataset itself		
P38										15:52:30 if based on the use case or user reviews, I determine this dataset could be useful, then look into documentation	procedural	
P38							D			15:53:23 expect to get data dictionary from documentation: the meaning of each feature		
P38										15:55:36 more curious about the data	I just more curious about the data itself.	
P38							T			15:56:35 good to see the sample, maybe just top few rows	assume sample data is the first few rows	
P38							P			15:56:49 maybe want to see a quick summary of the data-then profiling tool		
P38							U			15:58:21 check the use cases first to see if it has been used for similar purpose		
P38										15:58:31 then spend time reviewing user reviews		
P38		Y					U			15:58:45 data usage is case by case, see sample data may not be sufficient for determining whe the user review and the ratings give me more information about how different people use this data set for different case		
P38							D			15:59:33 if believe could use it for something, then go to documentation	procedural	
P38							U			16:01:22 use cases are one or two sentence description of the dataset's intended usage scenarios: fraud detection		
P38				Y			R			16:01:34 user reviews: the dataset is good or bad, quality problems		
P38				Y						16:02:09 not too many missing values & have many features		
P38		Y								16:03:10 not going to leave reviews if dataset is good		
P38							U			16:04:26 expect it to be provided by the data provider or other users		
P38										16:05:19 may look into more if the use cases seems somewhat relevant		
P38		Y		Y			D			16:06:01 data coverage can also play a role here: if too small, then probably too niche to be relevant		
P38							C, X			16:07:51 60% features selected based on peer review experience or domain knowledge		
P38				Y						16:08:01 another step before using the data is to do some data quality check		
P38		Y	Y	Y			P			16:08:35 if too many missing values, then use other sets		
P38		Y	Y	Y			V, P			16:08:42 if okay, then visualize the features to check distributions, relationships between features, and relationships between features and target variables		
P38		Y	Y	Y						16:09:15 then switch to model building		
P38		Y	Y	Y			P			16:10:52 drop the features/datasets if important feature has too many missing values		
P38		Y	Y	Y						16:12:28 incorrect values: like a negative number for cell phone price		
P38		Y	Y	Y			V, P			16:14:20 how to check: visualizations, or get min, max, median values		
P38							S, V			16:14:44 sampling some data to plot		
P38							P, V			16:15:23 sometimes came about distributions of the data		
P38								Y		do some cost-benefit analysis, won't pay for it himself		
P38		Y					P, V			16:23:43 mentioned kaggle, some distributions, and sample data: usually the dataset doesn't have a lot of features		
P38		Y					P, V			16:28:35 do more analysis after seeing the result: iterate over the features		
P38		Y					P, V			usually won't check for every feature if there are too many features		
P38		Y					P, V			16:29:51 whether to check every feature depends on the task's urgency	constraints: tight deadline	
P38										16:31:41 use some visualization feature to rank the importance of columns and see contribution of each feature		
P38							V			16:32:34 explain your model to clients	visualization as presentation	
P34										11:03:38 most used data from internal platforms		
P34										11:04:04 looked data for personal project		
P34										11:04:22 use fitbit to track sleep health		
P34	P						R, X			11:04:27 search kaggle to see others' approaches	I would just search Kaggle like existing projects along that domain just to see what other professionals are doing and seeing their approaches	
P34										11:04:54 also see their notebooks and codes		
P34										11:05:46 worked on an athletics program		
P34										11:05:57 2 components: on season-work with athletes and coaches, off-season-work with finance		
P34	D									11:06:47 centralize multiple data sources		
P34										11:07:40 use data to price tickets, when to start game...		
P34										11:08:05 give another example with VR: how much they're being paid, how much they work...		
P34										11:08:29 not look for external data because approach the business functions on its own		
P34	Y									11:11:10 different from laige, data-driven company, just started to incorporate more data into decisions		
P34										11:11:21 work from scratch: create the dataset myself		
P34										11:12:08 consolidated database into various tables depends on the business problem		
P34							D			11:12:13 document how the table is structured, so it would be easy to be expanded on		
P34							C			11:12:28 after collection, go back to stakeholder and recheck with them about the problem they're trying to solve		
P34							C			11:12:43 do a lot of back and forth to help them with decision making		
P34							C			11:12:55 work with sports performance coaches, who know which parameter to look into		
P34							D			11:16:06 check metadata first to see if it fits the specification I expect	I would say, like immediately, I'm looking at metadata on the data set to see if it exactly fits the specifications of what I'm looking for.	
P34							D			11:16:22 the specification entails the types of data to work with	Because when I'm approaching a problem, I already know what type of data I'm trying to collect and model with once I fit the specifications.	
P34							D			documentation includes information on source reliability	I'm gonna look at to see if it's a valid data source to see if it's something I can trust, so I source reputation plays a role	
P34							T			11:16:42 will check sample table right away		
P34							F, X			11:16:48 relates usage metrics to dataset's updates and some tags (metadata) indicate whether it's from an official data source or not		
P34	Y									11:17:01 less likely to use dataset from unofficial source	I'm thinking this is the just a random data set, so I might not use it.	
P34							R			11:17:16 in work scenario, unlikely to check user ratings cause trust the data, but could be helpful might be useful to have user ratings. If it's for professional use, like for my job, I'm never looking at this because I'm already working with data that I know is trustworthy. It's something we collect. But yeah, for like a general use case, On Kaggle, that's what I would be looking at.		
P34										think it could be interchangeable to the medals previous mentioned		
P34										11:17:37 already have a task in mind, so personally wouldn't think use case helpful		
P34										11:19:13 favor reputable data provider		
P34	Y						D			11:19:19 only trust data from api call	I think there's various ways you can export data for fitness trackers. But I'm only gonna trust something that comes directly from an Api call	
P34							D, C			11:19:35 where it's come from it's important unless you know someone internally	As a general consumer, I think you're somewhat limited to just tracker Api data, unless you know someone internally	
P34							D			11:19:50 information about how the data's formatted and what it's supposed to solve comes after the source		
P34							D			11:20:43 documentation: where it was being sourced and what is contained		
P34										11:20:50 these info together gave an idea of what I can potentially to solve with it		
P34							D, S, T			11:21:20 first check what's actually in the dataset then look back to the task at hand		
P34	D									11:22:36 have question at hand, then search for the general domain. With candidate datasets, look into the source, and then the data to see if it can answer the question		
P34							P, S			11:23:30 perform some analysis (with a subset of data), then check again to see the data can be used for that purpose	some back and forth efforts might be involved	
P34	P									11:23:36 only open for kaggle since most mainly used kaggle in the past		
P34							R, X			11:23:43 check user ratings and others' posted notebooks to see if they approached the same way		
P34										11:23:58 after all these procedures, download the data and use it	procedural	
P34							S, T			11:24:28 both want to see sample table and export some data		
P34										11:25:15 perform evaluation before extensive use		
P34							D, T, S			11:25:19 rely on metadata: if source is trustworthy, is line with my approach, see the data myself and then do some analysis		
P34	Y						D, S			11:26:32 while working with data from unofficial source, dive into the data collection details: check if the dataset is well-balanced, amount of missing values, and whether can be used for modeling		

